

Variably Saturated Flow in MODFLOW 6

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August 6, 2026

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1 Theoretical Development

In this section ...

1.1 Conservation of Mass

In case of variably saturated flow, the void space is occupied by two fluid phases, air and water, and a full description includes the simultaneous flow of both phases. For the process described here it is assumed that there is no mass transfer between the phases and the air is stagnant such that its pressure equals zero everywhere. Under those conditions, the differential macroscopic mass balance equation for the water phase has the form (c.f. equation (6.3.3) in Bear and Cheng, 2010):

$$\frac{\partial(\rho s \phi)}{\partial t} + \nabla \cdot (\rho \mathbf{q}) - \rho \Gamma = 0, \quad (1)$$

where

- ρ = liquid density, [ML⁻³]
- s = liquid saturation, [-]
- ϕ = porosity, [-]
- t = time, [T]
- $\mathbf{q}$ = vector of liquid flux per unit area, [LT⁻¹]
- Γ = volumetric source-sink term accounting for liquid added to ($\Gamma > 0$) or taken away ($-\Gamma < 0$) from the volume Ω , per unit volume per unit time, [T⁻¹]

Integrating equation 1 over a volume of porous medium V bounded by a surface σ , and applying Gauss’s theorem leads to:

$$\int_V \frac{\partial(\rho s \phi)}{\partial t} dV + \oint_{\sigma} \rho \mathbf{q} \cdot \mathbf{n} d\sigma - \int_V \rho \Gamma dV = 0, \quad (2)$$

where $\mathbf{n}$ equals the normal vector to the surface σ . Equation 2 states that the rate of change of mass stored in V must be balanced by the sum of liquid flux across the surface boundary of V and of liquid added by sources or removed at sinks.

The volume V is assumed small enough that within it, the liquid density (ρ), saturation (s), and porosity (ϕ) can be considered constant "representative" values, such that the first term of equation 2 can be expressed as

$$\int_V \frac{\partial(\rho s \phi)}{\partial t} dV = V \frac{\partial(\rho s \phi)}{\partial t}, \quad (3)$$

and the third term as

$$\int_V \rho \Gamma dV = \rho \Gamma V. \quad (4)$$

Equation 2 now becomes

$$V \frac{\partial(\rho s \phi)}{\partial t} + \oint_{\sigma} \rho \mathbf{q} \cdot \mathbf{n} d\sigma - \rho \Gamma V = 0. \quad (5)$$

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NOTE by AMP: Looks like the density doesn't divide out the way we'd hoped. Start with eqn 2, rewritten so that we're distinguishing between ρ , the uniform density in the cell, ρ_{σ} , the density associated with inflows and outflows along the boundary of the cell, and ρ_{Γ} , the density associated with other sources and sinks to and from the cell. Then, recognizing that ρ , s , and ϕ are uniform throughout the cell volume V ,

$$\int_V \frac{\partial(\rho s \phi)}{\partial t} dV + \oint_{\sigma} \rho_{\sigma} \mathbf{q} \cdot \mathbf{n} d\sigma - \int_V \rho_{\Gamma} \Gamma dV = 0, \quad (6)$$

$$\frac{\partial}{\partial t} \int_V \rho s \phi dV + \rho \oint_{\sigma} \frac{\rho_{\sigma}}{\rho} \mathbf{q} \cdot \mathbf{n} d\sigma - \rho \int_V \frac{\rho_{\Gamma}}{\rho} \Gamma dV = 0, \quad (7)$$

$$V \frac{\partial}{\partial t} (\rho s \phi) + \rho \oint_{\sigma} \frac{\rho_{\sigma}}{\rho} \mathbf{q} \cdot \mathbf{n} d\sigma - \rho \int_V \frac{\rho_{\Gamma}}{\rho} \Gamma dV = 0, \quad (8)$$

$$\rho s S_s V \frac{\partial h}{\partial t} + \rho \phi V \frac{\partial s}{\partial t} + \rho \oint_{\sigma} \frac{\rho_{\sigma}}{\rho} \mathbf{q} \cdot \mathbf{n} d\sigma - \rho \int_V \frac{\rho_{\Gamma}}{\rho} \Gamma dV = 0, \quad (9)$$

$$s S_s V \frac{\partial h}{\partial t} + \phi V \frac{\partial s}{\partial t} + \oint_{\sigma} \frac{\rho_{\sigma}}{\rho} \mathbf{q} \cdot \mathbf{n} d\sigma - \int_V \frac{\rho_{\Gamma}}{\rho} \Gamma dV = 0. \quad (10)$$

The resulting balance equation shows that to conserve mass, we have to scale the boundary and source-sink flows by appropriate density ratios. This is essentially using "apparent volumes." In other words, $\frac{\rho_{\sigma}}{\rho} \mathbf{q}$ and $\frac{\rho_{\Gamma}}{\rho} \Gamma$ are the volumetric flow rates for water at density ρ that correspond to the same *mass* flow rates as the

actual volumetric flow rates $\mathbf{q}$ (at density ρ_σ) and Γ (at density ρ_Γ). Seems this concept is bound to come up (in some form) if we want to use a volume balance and still conserve mass in some sense.

Another way to look at it is that for eqn 10 to reduce to a straight-up volume balance, we need $\rho_\sigma = \rho_\Gamma = \rho$. That implies that all the incoming flows – whether from other cells (or model features), or from external sources – need to be at density ρ . Propagating this requirement from cell to cell¹, we end up requiring that all flows throughout the model be at the same density. That corresponds to our “constant-density flow” assumption with variations in density being confined to specific storage, which is the idea behind apparent volume.

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REPLY by MJR: I think you are right that equation 2 complicates matters. The density does not divide out, as you mention, but I think it’s a result of performing the volume integral too soon here. Let’s keep working with the form from equation 1 a little bit longer. The divergence term can be expanded as

$$\nabla \cdot (\rho \mathbf{q}) = \mathbf{q} \cdot (\nabla \rho) + \rho \nabla \cdot \mathbf{q} \quad (11)$$

and the storage term contains a component

$$\frac{\partial(\rho s \phi)}{\partial t} = \dots + s \phi \frac{\partial \rho}{\partial t} \quad (12)$$

It is my impression that both in literature and in practice the term with $\nabla \rho$ is discarded assuming that the temporal variation of ρ is much greater than its spatial variation:

$$\left| s \phi \frac{\partial \rho}{\partial t} \right| \gg |\mathbf{q} \cdot (\nabla \rho)| \quad (13)$$

Let’s assume this is valid for now, then equation 1 becomes

$$\frac{\partial(\rho s \phi)}{\partial t} + \rho \nabla \cdot \mathbf{q} - \rho \Gamma = 0, \quad (14)$$

If we combine this with the storage term as given by equation 27, we arrive at the following macroscopic balance equation:

$$\rho s S_s \frac{\partial h}{\partial t} + \rho \phi \frac{\partial s}{\partial t} + \rho \nabla \cdot \mathbf{q} - \rho \Gamma = 0. \quad (15)$$

Here it seems that we can safely divide out the density. We have to keep in mind that Γ is a volumetric source/sink of water with a density equal to that of the water at the location where it is applied. In a strict sense this is not a valid mass source, but it is consistent with how things are done in MODFLOW 6.

After some reading and thinking, I believe that I understand why equation 13 holds true, but I look forward to discuss further. Anyway, say we pump into (or

¹Assuming there are no hydraulically disconnected parts of the model.

out of) an aquifer, this will change the pressure by a significant amount. The change will propagate through the water with the speed of sound: $1500m/s$. This combination should be compared to $\mathbf{q}\nabla\rho$ which represents how much the density changes at a fixed point because a fluid of a different density is physically flowing past it. In this case the velocity is extremely small, namely that of Darcy flow, say $10^{-5}m/s$. And for the problems we are considering here, the water is assumed homogeneous with no variation in solute concentration or temperature. This makes that $\nabla\rho$ itself is negligibly small.

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NOTE by AMP: Thanks for your response – it makes sense. In my analysis, getting to the volume balance hinges on the idea of apparent volume. In yours, it hinges on assuming that the spatial variation is negligible compared to the temporal variation. But I think they’re actually two sides of the same coin – two different ways of saying that we’re going to neglect density variations insofar as flows are concerned, but we’ll account for their effect on storage. Your analysis makes that assumption earlier on, before integration over the cell volume, whereas mine does so after the integration, but i think the underlying idea is the same. One thing to note is that in eqns 1 and 14 I think the density in $\rho\Gamma$ needs to be interpreted as the density of the source or sink, which for a source can be different than the prevailing density at that location in the aquifer. So the assumption of no spatial variation in density needs to be extended to external sources, as well, which is perfectly consistent with the assumptions made in my analysis. We should discuss, but at this point my feeling is that we’re essentially saying the same thing, but you’re working with differential volumes, and I’m working with non-infinitesimal volumes for which the assumptions being made are maybe easier to conceptualize.

=====

1.2 Fluid-Flux Equation

The fluid flux $\mathbf{q}$ is given by Darcy’s law for unsaturated flow:

$$\mathbf{q} = -\frac{\mathbf{k}(s)}{\mu} \cdot (\nabla p + \rho g \nabla z), \quad (16)$$

where the permeability of the medium $\mathbf{k}$ depends on its saturation s , μ is the dynamic viscosity, p the pressure of the fluid, and g the gravitational acceleration. For constant density and ignoring the temperature dependence of viscosity, this can be writtten in terms of hydraulic head h as

$$\mathbf{q} = -\mathbf{K}(s) \cdot \nabla h, \quad (17)$$

with $\mathbf{K}$ a rank-2 tensor of hydraulic conductivity. The hydraulic head equals the sum of the pressure head ψ and the vertical coordinate z :

$$h = \psi + z. \quad (18)$$

The concept of relative hydraulic conductivity² is commonly used to capture the dependence of the conductivity on saturation. For example, in an isotropic porous medium:

$$\mathbf{q} = -k_r(s)K_{\text{sat}}\nabla h, \quad (19)$$

where K_{sat} is the hydraulic conductivity tensor at full saturation, the relative hydraulic conductivity is given by:

$$k_r = \frac{K(s)}{K_{\text{sat}}}. \quad (20)$$

This can be easily extended to the anisotropic case if the principal axes of the conductivity tensor are aligned with the coordinate axes:

$$\mathbf{K}(s) = \mathbf{k}_r \cdot \mathbf{K}_{\text{sat}} = \begin{bmatrix} k_r^x & 0 & 0 \\ 0 & k_r^y & 0 \\ 0 & 0 & k_r^z \end{bmatrix} \cdot \begin{bmatrix} K_{\text{sat}}^x & 0 & 0 \\ 0 & K_{\text{sat}}^y & 0 \\ 0 & 0 & K_{\text{sat}}^z \end{bmatrix}, \quad (21)$$

although this poses the challenge to determine not one but three, in this case direction dependent, relative hydraulic conductivity functions. In case the medium is anisotropic and the axes are not aligned, the conductivity tensor follows from the previous expression by a rotation of coordinates:

$$\mathbf{K}(s) \rightarrow \mathbf{R}\mathbf{K}(s)\mathbf{R}^T = (\mathbf{R}\mathbf{k}_r\mathbf{R}^T) \cdot (\mathbf{R}\mathbf{K}_{\text{sat}}\mathbf{R}^T), \quad (22)$$

with $\mathbf{R}$ a general three-dimensional rotation matrix such that $\mathbf{R}\mathbf{R}^T = \mathbf{1}$. This can in principle be used to treat variably saturated flow in three-dimensional anisotropic media, consistent with the XT3D capability [7] available in MODFLOW 6 for saturated groundwater flow. For practical reasons, such as limiting the challenge of calibrating the conductivities in a model, it is assumed that relative hydraulic conductivity is isotropic. This implies that equation 17 can be written as

$$\mathbf{q} = -k_r(s)\mathbf{K}_{\text{sat}} \cdot \nabla h, \quad (23)$$

where k_r is a scalar function of saturation. This approximation has been used in many simulation codes (e.g., [5, 9, 11, 6]), but it has been pointed out that the effect of saturation generally depends on direction [1].

Substituting the expression for the fluid flux into equation 2 leads to

$$V \frac{\partial(\rho s \phi)}{\partial t} - \oint_{\sigma} \rho k_r(s) \mathbf{K}_{\text{sat}} \cdot \nabla h \cdot \mathbf{n} d\sigma - \rho \Gamma V = 0. \quad (24)$$

If the terms in the surface integral can be considered constant over each of the m faces of an arbitrary prismatic volume element (isotropic!?), this can be written as:

$$V \frac{\partial(\rho s \phi)}{\partial t} - \rho \sum_{\alpha=1}^m \overline{k_r K_{\text{sat}}} \left[\frac{\partial h}{\partial n} \right]_{\alpha} \sigma_{\alpha} - \rho \Gamma V = 0, \quad (25)$$

²More fundamental would be relative permeability, but we are not planning on using that here, do we?

where σ_α equals the area of the α face and the bar indicates that an effective value is used for hydraulic conductivity, to be discussed in more detail in section 2.2. Additionally, the head gradient is assumed to be normal to the faces bounding the volume which means that for any face α it follows that: $(\nabla h)_\alpha = (\partial h / \partial n)_\alpha \mathbf{n}_\alpha$.

1.3 Storage Term

The change in mass of water that is held in storage is given by the first term in equation 25. This can be expressed in terms of hydraulic head, the principal dependent variable for solving the system of equations, as follows:

$$\frac{\partial(\rho s \phi)}{\partial t} = \left[s \phi \frac{\partial \rho}{\partial h} + \rho s \frac{\partial \phi}{\partial h} \right] \frac{\partial h}{\partial t} + \rho \phi \frac{\partial s}{\partial t}. \quad (26)$$

Between brackets are two distinct terms contributing to the water added to or released from storage due to changes of the hydraulic head: the fluid compressibility term to account for changes in the fluid density ρ and a term to represent the (elastic) compression of the solid matrix given by the change in the porosity of the medium ϕ . The last term on the right accounts for the change in soil saturation, which is a function of pressure head ψ , or, in this case where the z coordinate is fixed, of hydraulic head: $s = s(h)$.

For any particular type of soil or geologic formation, these variables are assumed to be a function of pressure only. The following definitions are introduced to restructure the right-hand side of equation 26:

$$\text{Fluid compressibility coefficient: } \beta_c = \frac{1}{\rho} \frac{\partial \rho}{\partial p}$$

$$\text{Solid matrix compressibility coefficient: } \alpha_c = \frac{1}{1 - \phi} \frac{\partial \phi}{\partial p}.$$

Now with the approximation that the density of water depends on pressure only³, $\rho = \rho(p)$, the change in storage can be written as:

$$\frac{\partial(\rho s \phi)}{\partial t} = \rho s S_s \frac{\partial h}{\partial t} + \rho \phi \frac{\partial s}{\partial t}. \quad (27)$$

The specific storage⁴ in this equation is defined as

$$S_s = \rho g [\phi \beta_c + (1 - \phi) \alpha_c]. \quad (28)$$

³Not seeing how this works now, will need to review. I think it is in the spirit of Bear and Cheng, 5.1.31 and 5.1.32

⁴It should be noted that the assumptions underlying the specific storage formulation don't necessarily hold in the unsaturated case, but the error from including it is generally small; see, for example, the SUTRA manual.

1.4 Variably Saturated Flow Equation

Putting the results from the previous sections together, substituting the storage terms (equation 27) into the differential form of the balance law (equation 25), leads to Richards' equation, a nonlinear partial differential equation for variably saturated flow:

$$V \left(s S_s \frac{\partial h}{\partial t} + \phi \frac{\partial s}{\partial t} \right) - \sum_{\alpha=1}^m \overline{k_r K_{\text{sat}}} \left[\frac{\partial h}{\partial n} \right]_{\alpha} \sigma_{\alpha} - \Gamma V = 0. \quad (29)$$

with (repeated here for convenience):

V	=	the representative elementary volume
s	=	water saturation
S_s	=	specific storage
h	=	hydraulic head
ϕ	=	porosity,
$\overline{k_r}$	=	effective relative hydraulic conductivity
$\overline{K}$	=	effective hydraulic conductivity
$(\partial h / \partial n)_{\alpha}$	=	head gradient normal to face α
σ_{α}	=	surface area of face α
Γ	=	volumetric source-sink term

During the assembly of equation 29, the density ρ appears as a factor in all the terms and cancels out. This marks the transition between the mass based balance law that is the starting point of the derivation, to a formulation that rests on a conservation of water volume as used in MODFLOW 6. Note that as the medium approaches full saturation, the derivative $\partial s / \partial t$ vanishes and k_r becomes equal to 1. In that case, the above expression converges smoothly to the saturated groundwater flow equation used in MODFLOW 6:

$$V S_s \frac{\partial h}{\partial t} - \sum_{\alpha=1}^m \overline{K_{\text{sat}}} \left[\frac{\partial h}{\partial n} \right]_{\alpha} \sigma_{\alpha} - \Gamma V = 0. \quad (30)$$

It is this concept of continuity that is used to integrate the variably saturated flow mechanism into the existing flow formulation.

1.5 Initial and Boundary Conditions

1.6 Analytical Functions for Soil Properties

1.6.1 Water Retention Curve

The water retention curve, also known as the soil moisture characteristic, describes the relation between the degree of saturation and the pressure potential

of the water in the soil. As the capillary pressure becomes larger, and the pressure potential becomes increasingly negative, the water saturation reduces up to a residual moisture content. In what follows the water retention curve will be given for a number of models through an equation for the effective water saturation s_e , sometimes called a reduced water saturation, which is defined as:

$$s_e = \frac{s - s_r}{1 - s_r} \quad (31)$$

where the fitting parameters s_r is the irreducible (or residual) water saturation ($0 < s_r < 1$).

These characteristic curves are generally determined from experiment by fitting a parametrized algebraic function to the measured data. A benefit of using such a function is that its derivative can be used to define the specific moisture capacity. Additionally, integration leads to a relation between relative hydraulic conductivity and pressure potential [?]. These three analytical functions are required for the numerical solution of equation 29. Figure ?? shows experimental data and their fits for different types of soil.

Brooks-Corey The Brooks-Corey model [2] gives the soil moisture characteristic in terms of effective saturation s_e as:

$$s_e(\psi) = \begin{cases} \left(\frac{\psi_b}{\psi}\right)^\lambda & \psi \leq \psi_b \\ 1 & \psi > \psi_b \end{cases} \quad (32)$$

with the parameters

$$\begin{aligned} \psi_b &= \text{bubbling or air entry pressure potential } (\psi_b < 0) \\ \lambda &= \text{Brooks-Corey exponent } (\lambda > 0) \end{aligned}$$

The parameter ψ_b is equal to the pressure potential required to desaturate the largest pores in the medium. The exponent λ is the pore size distribution index. Both parameters are characteristics of the medium.

Haverkamp The following soil moisture characteristic has been used in a study by Haverkamp et al. [4]:

$$s_e(\psi) = \begin{cases} \frac{1}{1 + (-\alpha\psi)^n} & \psi \leq 0 \\ 1 & \psi > 0 \end{cases} \quad (33)$$

where the parameters α and n are both positive. When comparing this to Brooks-Corey for large values of $-\psi$, it becomes clear that α is inversely related to the air entry pressure potential ψ_b .

Van Genuchten The moisture retention function by Van Genuchten [10] is defined as

$$s_e(\psi) = \begin{cases} \frac{1}{(1 + (-\alpha\psi)^n)^m} & \psi \leq 0 \\ 1 & \psi > 0 \end{cases} \quad (34)$$

where α , m , and n are the Van Genuchten fitting parameters, all positive and with $m = 1 - 1/n$. For large values of n (when $m \rightarrow 1$) the function will behave similarly to the expression given by Haverkamp above.

1.6.2 Specific Moisture Capacity

The specific moisture capacity function describes the change in water content θ resulting from a unit change in pressure head ψ . It is related to the derivative of the water retention curve discussed above:

$$C_m(\psi) = \frac{\partial\theta}{\partial\psi} \approx \phi \frac{\partial s}{\partial\psi} \quad (35)$$

where it is assumed that changes in s are much larger than changes in ϕ . The moisture capacity will appear as the dominant component of the storage coefficient for variably saturated flow, either through the analytical expression of $C_m(\psi)$ or by a discrete approximation (chord-slope) of the derivative.

Brooks-Corey Converting from s_e to s in equation 32 and then taking the derivative leads to the Brooks-Corey expression for the specific moisture capacity function:

$$C_m(\psi) = \begin{cases} -\phi(1 - s_r) \frac{\lambda}{\psi_b} \left(\frac{\psi_b}{\psi} \right)^{\lambda+1} & \psi \leq \psi_b \\ 0 & \psi > \psi_b \end{cases} \quad (36)$$

Note that this function is discontinuous at $\psi = \psi_b$. Also $\psi_b < 0$ which means that $C_m \geq 0$ at all times: saturation is a monotonously increasing function of pressure head.

Haverkamp When the soil moisture characteristic is represented by the Haverkamp parametrization, the following equation is used:

$$C_m(\psi) = \begin{cases} \phi(1 - s_r) \frac{n\alpha(-\alpha\psi)^{n-1}}{(1 + (-\alpha\psi)^n)^2} & \psi \leq 0 \\ 0 & \psi > 0 \end{cases} \quad (37)$$

Van Genuchten

$$C_m(\psi) = \begin{cases} \phi(1 - s_r) \frac{mn\alpha(-\alpha\psi)^{n-1}}{(1 + (-\alpha\psi)^n)^{m+1}} & \psi \leq 0 \\ 0 & \psi > 0 \end{cases} \quad (38)$$

1.6.3 Relative Permeability

Brooks-Corey

$$k_r(\psi) = \begin{cases} \left(\frac{\psi_b}{\psi}\right)^{2+3\lambda} & \psi \leq \psi_b \\ 1 & \psi > \psi_b \end{cases} \quad (39)$$

Haverkamp In practice, for the Haverkamp model, the following equation is used to fit to the measured hydraulic conductivities at varying saturation:

$$k_r(\psi) = \begin{cases} \frac{1}{1 + (-\beta\psi)^k} & \psi \leq 0 \\ 1 & \psi > 0 \end{cases} \quad (40)$$

Although this is exactly the same type of function as given for the soil moisture in equation 33, the value of the coefficients β and k are not the same.

Van Genuchten

$$k_r(\psi) = \begin{cases} \frac{\left[\frac{(1 - (-\alpha\psi)^{n-1}) (1 + (-\alpha\psi)^n)^{-m}}{[1 + (-\alpha\psi)^n]^{m/2}} \right]^2}{1} & \psi \leq 0 \\ 1 & \psi > 0 \end{cases} \quad (41)$$

where α , n , and m are the Van Genuchten fitting parameters introduced in the expression for the soil moisture characteristic and $m = 1 - 1/n$ as before.

1.6.4 Hysteresis

2 Numerical Solution

Equation 29, together with the boundary conditions as described in section 1.5, is a highly non-linear partial differential equation that has no general closed-form solution. Although it can be considered a more advanced description of unconfined and partially saturated groundwater flow than the standard formulation available in MODFLOW 6, the solution strategy is the same. The variably saturated flow equation is discretized using a control-volume finite-difference (CVFD) method. The method replaces the continuum of coordinates in equation 29 with a discrete set of points in space and time. The control volume is called a cell and the point at the center of the cell a node. The collection of cells with their connections is the model grid. The partial derivatives in the equation can be approximated by differencing head values between neighbouring cells and between subsequent time values. The resulting finite-difference equations will have terms that are a function of the discretized dependent variable, such as $k_r = k_r(h)$ and $s = s(h)$. These non-linearities can be overcome

using a standard Picard iteration method⁵ or by applying the Newton-Raphson linearization that is known to provide more stability when solving for cells that are under a water-table condition. For each cell in the the grid that is under variably saturated flow conditions, the linearized balance terms are added to the system representing the motion equations in the total (saturated and unsaturated) simulation domain. This system is solved by the MODFLOW iterative solver as part of its outer (non-linear) iteration loop.

Water Balance on a Cell

The variably saturated flow equation in CVFD form can be used to express the concept of water balance on a cell n as

$$\sum_{m \in \eta_n} Q_{n,m} + Q_{n,\Gamma} - Q_{n,\Sigma} = 0, \quad (42)$$

where

$$\begin{aligned} Q_{n,m} &= \text{flow rate from cell } m \text{ into } n \\ \eta_n &= \text{collection of cells connected to } n \\ Q_{n,\Gamma} &= \text{summed contribution from sources and sinks in the cell} \\ Q_{n,\Sigma} &= \text{change in volume of water stored in } n \text{ per unit time} \end{aligned}$$

and the terms in equation 42 are all of dimension $[\text{L}^3\text{T}^{-1}]$. For every cell n , after reordering, this water balance equation corresponds to the n^{th} row in the matrix equation

$$\mathbf{A}\mathbf{h} = \mathbf{b}, \quad (43)$$

where $\mathbf{A}$ equals a $n \times n$ coefficient matrix, $\mathbf{h}$ the dependent variable vector at the end of the time step, and $\mathbf{b}$ the constant terms grouped in a vector that is commonly referred to as the right-hand side.

For general variably saturated flow problems, the coefficient matrix is non-linear and depends on the head values that are to be calculated. Using the Picard iteration method, equation 43 can be written as

$$\bar{\mathbf{A}}^{(k-1)}\mathbf{h}^{(k)} = \mathbf{b}^{(k-1)}, \quad (44)$$

such that $\mathbf{h}^{(k)}$ is the head vector at the new iteration level k and $\bar{\mathbf{A}}^{(k-1)}$ is the linearized coefficient matrix that no longer depends on $\mathbf{h}^{(k)}$ but is evaluated using values from the previous iteration level $k - 1$ instead, and likewise for $\mathbf{b}^{(k-1)}$. Repeatedly formulating and solving the system until $\mathbf{h}^{(k)} \approx \mathbf{h}^{(k-1)}$ produces the numerical solution of equation 43.

In the following sections, discretized expressions are presented for all the terms in equation 42. These expressions, together with the linearization procedure and a discrete formulation of available initial conditions and boundary conditions, are required to numerically solve the process of variably saturated flow within the MODFLOW 6 framework.

⁵Are we planning to support non-NR treatment of Richards flow?? AMP: good question - was wondering about that, too

Newton-Raphson Linearization

The Newton-Raphson method is an alternative way of linearizing the balance equation. First define the residual vector $\mathbf{r}$ for a cell n using the component notation as

$$r_n = \sum_{m \in \eta_n} Q_{n,m} + Q_{n,\Gamma} - Q_{n,\Sigma}. \quad (45)$$

The coefficients of the Jacobian matrix $\mathbf{J}$ are given by

$$J_{n,r} = \frac{\partial r_n}{\partial h_r}. \quad (46)$$

The strategy is to find the change in head $\Delta \mathbf{h}$ that drives the residual vector to zero. This can be expressed with the following iterative scheme:

$$\mathbf{J}^{(k-1)} \Delta \mathbf{h}^{(k)} = -\mathbf{r}^{(k-1)}. \quad (47)$$

Rewriting this in a form similar to equation 44 leads to

$$\mathbf{J}^{(k-1)} \mathbf{h}^{(k)} = -\mathbf{r}^{(k-1)} + \mathbf{J}^{(k-1)} \mathbf{h}^{(k-1)}. \quad (48)$$

The last expression explicitly depends on head instead of the head upgrade and makes it easier to solve the problem with a unified implementation for the standard formulation and the Newton-Raphson method.

Combining equations 45, 46, and 48, and separating the diagonal and off-diagonal part of $\mathbf{J}$, leads to the full CVFD equation for the Newton-Raphson formulation:

$$\begin{aligned} & \left(\sum_{m \in \eta_n} \frac{\partial Q_{n,m}}{\partial h_n} + \frac{\partial Q_{n,\Gamma}}{\partial h_n} - \frac{\partial Q_{n,\Sigma}}{\partial h_n} \right) h_n^{(k)} + \sum_{m \in \eta_n} \frac{\partial Q_{n,m}}{\partial h_m} h_m^{(k)} = \\ & - \left(\sum_{m \in \eta_n} Q_{n,m} + Q_{n,\Gamma} - Q_{n,\Sigma} \right) + \left(\sum_{m \in \eta_n} \frac{\partial Q_{n,m}}{\partial h_n} + \frac{\partial Q_{n,\Gamma}}{\partial h_n} - \frac{\partial Q_{n,\Sigma}}{\partial h_n} \right) h_n^{(k-1)} \\ & + \sum_{m \in \eta_n} \frac{\partial Q_{n,m}}{\partial h_m} h_m^{(k-1)}, \end{aligned} \quad (49)$$

where all coefficients are to be evaluated at iteration $k-1$. This is the expression given in the documentation for the MODFLOW 6 groundwater flow model and is repeated here for convenience. The following sections will detail the various contributions to the terms in this equation.

2.1 Spatial Discretization and Internal Flows

2.2 Intercell Averaging of Conductance Terms

Saturated Hydraulic Conductivity

Relative Hydraulic Conductivity

2.3 Temporal Discretization and Storage

For transient simulations, the change in storage is given by equation 27. Because of the presence of both h and s as unknowns, this expression is known as the mixed-form representation of Richards' equation. To proceed with the numerical solution, the time variable is discretized and a discrete representation of this term is required. A great deal of research has been concerned with how to best discretize the storage term such that it maintains the mass conservation that is inherent to the equation in the continuous domain. Applying the chain rule of differentiation to convert the time derivative of saturation to hydraulic head, i.e., making the substitution:

$$\frac{\partial s}{\partial t} \rightarrow \frac{\partial s}{\partial h} \frac{\partial h}{\partial t}, \quad (50)$$

will generally lead to mass balance errors in case the analytical (or tangent) derivative for $\partial s / \partial h$ is used. A more detailed discussion of the subject can be found in [8] and references therein.

Therefore, instead of applying the product rule in the continuous domain and working with the expression on the right-hand side of equation 27, discrete differences are used as a starting point. The volumetric rate of change over a time interval Δt in a representative elementary volume V is given by

$$Q_{\Sigma} = \frac{V \Delta(\rho s \phi)}{\rho \Delta t}. \quad (51)$$

The change in the mass of water stored per volume of aquifer can be expanded as

$$\Delta(\rho s \phi) \equiv (\rho s \phi)^i - (\rho s \phi)^{i-1} = (\rho \phi)^i s^i - (\rho \phi)^{i-1} s^{i-1}, \quad (52)$$

where i and $i - 1$ label the time levels at which terms are evaluated and the time interval Δt for which the change is determined, is given as

$$\Delta t = t^i - t^{i-1}. \quad (53)$$

Respecting equation 52 is clearly necessary to conserve mass, but not sufficient. The evaluation of the individual terms as so called state functions is essential and deserves particular attention when converting the dependent variable to head.

Some remarks: from here on I'd like to derive the discretized storage terms. I am not sure though if the splitting into two distinct storage rates (specific and

'unsaturated') is reasonable. And maybe there is no need for it anyway... Can we define Q_{S_s} as the rate that comes from varying h exclusively? The following three storage formulations from our notes/derivations are equivalent:

$$Q_{\Sigma} = \rho_0 \phi_0 \Delta s + S_s (\psi^i s^i - \psi^{i-1} s^{i-1}) \quad (54)$$

$$Q_{\Sigma} = \rho^{i-1} \phi^{i-1} \Delta s + S_s s^i \Delta h \quad (55)$$

$$Q_{\Sigma} = [\rho \phi]_{h=0} \Delta s + S_s (h^i s^i - h^{i-1} s^{i-1}) \quad (56)$$

So is it the second equation maybe, that naturally defines the two distinct rates.

Following the existing procedure in MODFLOW 6 for confined and unconfined groundwater storage, the general equation for storage contributions to the variably saturated flow equation is:

$$Q_{\Sigma} = Q_{\Delta s} + Q_{S_s}, \quad (57)$$

where Q_{S_s} is the volumetric flow rate associated with specific storage and $Q_{\Delta s}$ the volumetric flow rate resulting from changes in the water saturation, referred to in this document as unsaturated storage⁶.

Specific Storage

The conversion of $(\rho \phi)^i$ into an expression that depends on h^i , the hydraulic head at the new time level, starts with the approximating relation:

$$(\rho \phi)^i = (\rho \phi)^{i-1} + \frac{\Delta(\rho \phi)}{\Delta h} \Delta h \approx (\rho \phi)^{i-1} + \rho_o S_s \Delta h, \quad (58)$$

with ρ_o a constant value for the density⁷ and S_s , the specific storage, assumed to be constant over the discretized volume too. Substituting in equation 52 leads to:

$$\Delta(\rho s \phi) = (\rho \phi)^{i-1} \Delta s + \rho_o S_s s^i \Delta h. \quad (59)$$

After using a backward Euler scheme to approximate the time derivative, the discretized volumetric flow rate from specific storage changes for cell n equals:

$$Q_{n,S_s} = \rho_o S_s s_n^i \frac{h_n^i - h_n^{i-1}}{\Delta t}. \quad (60)$$

In the iterative scheme the current iteration is labeled by k , the expression is then linearized by evaluating the saturation at the previous iteration level, with the replacement $s_n^{i,(k)} \rightarrow s_n^{i,(k-1)}$:

$$Q_{n,S_s} = \rho_o S_s s_n^{i,(k-1)} \frac{h_n^{i,(k)} - h_n^{i-1}}{\Delta t}. \quad (61)$$

⁶I don't think this is a correct term, but for now let's work with it. AMP: agree

⁷@AMP: I feel we need to go over our work carefully to remind ourselves where and how we assume densities to be constant for now. AMP: agree

Note that alternatively, equation 58 could have been used to express $(\rho\phi)^{i-1}$ explicitly in terms of $(\rho\phi)^i$ and substitute in equation 52. In that case, $Q_{n, Ss}$ would end up being proportional to s^{i-1} instead.⁸

To convert the term containing Δs in equation 59, two different approaches are available and worked out below: a chord slope method discussed in [8] and what is generally known as the modified Picard method [3].

Unsaturated Storage: Chord Slope Method

The conversion of Δs in equation 59 to a quantity that depends on head, requires some care. A straightforward discretization of the chain rule expansion, shown in equation 50, such that

$$\Delta s = \left. \frac{\partial s}{\partial h} \right|_{h^{i-\frac{1}{2}}} \Delta h, \quad (62)$$

leaves the question open on what intermediate time $t^{i-\frac{1}{2}}$ (or for what intermediate head value $h^{i-\frac{1}{2}}$) the derivative should be evaluated. Since in practice this derivative is highly non-linear, the resulting expression will generally lead to mass balance errors.

The approach used here is to move the chain rule to the discrete domain, starting with the trivial equality:

$$\Delta s = \left(\frac{\Delta s}{\Delta h} \right) \Delta h = \left(\frac{s^i - s^{i-1}}{h^i - h^{i-1}} \right) (h^i - h^{i-1}). \quad (63)$$

After division by Δt and a linearization that follows from evaluating the quotient at the previous iteration level, the discrete unsaturated storage term is given by:

$$Q_{n, \Delta s} = (\rho\phi)^{i-1} \left(\frac{s_n^{i, (k-1)} - s_n^{i-1}}{h_n^{i, (k-1)} - h_n^{i-1}} \right) \frac{h_n^{i, (k)} - h_n^{i-1}}{\Delta t}. \quad (64)$$

When convergence of the iterative scheme is achieved, or, put differently, when $h^{i, (k)} \approx h^{i, (k-1)}$, this relation turns into a state function of saturation and will therefore be mass conservative.

Unsaturated Storage: Modified Picard Scheme

An common method to introduce hydraulic head as the dependent variable for the term containing Δs in equation 59, is known as the modified Picard scheme [3]. The saturation at the new time level is approximated by a truncated Taylor series as:

$$s^i \approx s^{i, (k-1)} + C^{i, (k-1)} \left(h^{i, (k)} - h^{i, (k-1)} \right). \quad (65)$$

⁸The implications of which are still to be sorted out, maybe something for the appendix. AMP: agree

The moisture capacity function is defined in this equation by⁹:

$$C^{i,(k-1)} = \left. \frac{\partial s}{\partial h} \right|_{h^{i,(k-1)}}, \quad (66)$$

which in practice will be a different function depending on the type of soil in a cell. With this approximation, the volumetric flow rate for cell n associated with changes in saturation s_n is given by:

$$Q_{n,\Delta s} = (\rho\phi)^{i-1} \left(\frac{s_n^{i,(k-1)} - s_n^{i-1}}{\Delta t} + C_n^{i,(k-1)} \frac{h_n^{i,(k)} - h_n^{i,(k-1)}}{\Delta t} \right). \quad (67)$$

When the method converges, the term proportional to the moisture capacity function vanishes and the rate becomes identical to the one from the chord slope method above, thereby satisfying the mass conservative property as intended.

Total Storage: Newton-Raphson Scheme

The discrete change in total storage in equation 59 is the starting point for the Newton-Raphson linearization procedure. Dividing both sides by Δt gives for the volumetric flow rate associated with the total storage:

$$Q_\Sigma = \frac{\Delta(\rho s\phi)}{\Delta t} = (\rho\phi)^{i-1} \frac{s^i - s^{i-1}}{\Delta t} + \rho_o S_s s^i \frac{h^i - h^{i-1}}{\Delta t}. \quad (68)$$

The terms associated with the residual as written in the expansion in equation 49, for a cell n , are now given by:

$$Q_{n,\Sigma}^{(k-1)} = (\rho\phi)^{i-1} \frac{s_n^{i,(k-1)} - s_n^{i-1}}{\Delta t} + \rho_o S_s s_n^{i,(k-1)} \frac{h_n^{i,(k-1)} - h_n^{i-1}}{\Delta t}, \quad (69)$$

The Jacobian terms follow directly from taking the derivative of equation 68 with respect to h :

$$\frac{\partial Q_\Sigma}{\partial h} = \frac{(\rho\phi)^{i-1}}{\Delta t} \frac{\partial s}{\partial h} + \frac{\rho_o S_s}{\Delta t} \left(s^i + h^i \frac{\partial s}{\partial h} \right). \quad (70)$$

Finally, this needs to be determined at the previous iteration level $k-1$, which means evaluating at $h = h^{i,(k-1)}$. For an arbitrary cell n this leads to the following equation:

$$\begin{aligned} \left(\frac{\partial Q_{n,\Sigma}}{\partial h_n} \right)^{(k-1)} &= \frac{(\rho\phi)^{i-1}}{\Delta t} C_n^{i,(k-1)} \\ &+ \frac{\rho_o S_s}{\Delta t} \left(s_n^{i,(k-1)} + \left(h_n^{i,(k-1)} - h_n^{i-1} \right) C_n^{i,(k-1)} \right). \end{aligned} \quad (71)$$

Equations 69 and 71 are used to formulate the system of equations for the variably saturated flow process in case of a simulation with the Newton-Raphson linearization enabled.

⁹This is different from the function defined in eqn 35.

2.4 Sources and Sinks

2.5 Initial and Boundary Conditions

2.6 Adaptive Timestepping

3 Integration into MODFLOW 6

The capability presented in this report offers an alternative flow formulation in the NPF package for cells that are under variably saturated conditions. The storage package (STO) has been extended to include a more sophisticated treatment of partially saturated cells. New packages have been developed to account for infiltration (INF), ponding (PON), ... The existing packages for ... have been modified .

3.1 Infiltration

3.2 Seepage

3.3 Ponding

4 Model Verification

4.1 Analytical Results

4.2 Example Problems

A Ft. Lauderdale papers: detailed analysis of the storage formulation

Underlying the Newton formulation is the application of the following linear approximations in equation 59 (or, equivalently, equation 68):

$$\Delta s \approx \Delta s^{i,(k-1)} + \left. \frac{\partial s}{\partial h} \right|_{h^{i,(k-1)}} \left(h^{i,(k)} - h^{i,(k-1)} \right) \quad (72)$$

in the unconfined term, and

$$\begin{aligned} s^i \Delta h &\approx s^{i,(k-1)} \Delta h^{i,(k-1)} + \left. \frac{\partial (s \Delta h)}{\partial h} \right|_{h^{i,(k-1)}} \left(h^{i,(k)} - h^{i,(k-1)} \right) \\ &= s^{i,(k-1)} \Delta h^{i,(k)} + \left. \frac{\partial s}{\partial h} \right|_{h^{i,(k-1)}} \Delta h^{i,(k-1)} \left(h^{i,(k)} - h^{i,(k-1)} \right) \end{aligned} \quad (73)$$

in the confined term. Celia's method uses equations 65 and 66, which together are mathematically equivalent¹⁰ to equation 72, in the unconfined term and equation 61 in the confined term. Thus, we see that Celia essentially applies

¹⁰Actually, Celia linearizes $s\phi$, not s .

Newton's method to the unconfined term, but not the confined term. The chord-slope method, which uses equations 64 and 61, obviously does not apply Newton's method to either term. Replacing our chord slope with $\frac{s^{i,(k-1)} - s^{i,(k-2)}}{h^{i,(k-1)} - h^{i,(k-2)}}$ would yield an "unconfined secant method." Replacing it with the exact derivative evaluated at the previous iterate would yield Celia's "unconfined Newton's method."

The total-storage-change update from one iteration to the next can be written as

$$\Delta(\rho s \phi)^{i,(k)} = \Delta(\rho s \phi)^{i,(k-1)} + J^{est} \left(h^{i,(k)} - h^{i,(k-1)} \right), \quad (74)$$

where J^{est} is an estimate of the Jacobian. For the Newton formulation, substitution of equations 72 and 73 into equation 59 (or equation 68) and rearrangement leads to

$$J^{est} = (\rho \phi)^{i-1} \left. \frac{\partial s}{\partial h} \right|_{h^{i,(k-1)}} + \rho_o S_s s^{i,*}, \quad (75)$$

where

$$s^{i,*} \equiv s^{i,(k-1)} + \left. \frac{\partial s}{\partial h} \right|_{h^{i,(k-1)}} \Delta h^{i,(k-1)}. \quad (76)$$

As one would expect, this is exactly the Jacobian given by equation 71. For Celia's method,

$$J^{est} = (\rho \phi)^{i-1} \left. \frac{\partial s}{\partial h} \right|_{h^{i,(k-1)}} + \rho_o S_s s^{i,(k-1)}. \quad (77)$$

For the chord-slope method,

$$J^{est} = (\rho \phi)^{i-1} \left(\frac{s_n^{i,(k-1)} - s_n^{i-1}}{h_n^{i,(k-1)} - h_n^{i-1}} \right) + \rho_o S_s s^{i,k-1}. \quad (78)$$

B Comparison of different formulations of the storage term

Expansion of the storage term using the product rule gives

$$\frac{\partial(\rho s \phi)}{\partial t} = \rho \phi \frac{\partial s}{\partial t} + \rho s \frac{\partial \phi}{\partial t} + \phi s \frac{\partial \rho}{\partial t}. \quad (79)$$

The three terms on the right-hand side represent three ways to change the mass of water stored. The first term involves changing the volume of water by filling the existing pore space with more water of the same density. The second term involves changing the volume of pore space, adding or withdrawing water of the same density to maintain the same saturation. The third term involves changing the density of the water, keeping the same volume of water. The second and third terms on the right-hand side together represent specific storage.

B.1 Formulation in terms of volumetric water content

As an alternative, one can separate out only the density-derivative term, leaving the product $s\phi$, the volumetric water content, intact as a dependent variable:

$$\frac{\partial(\rho s\phi)}{\partial t} = \rho \frac{\partial(s\phi)}{\partial t} + \phi s \frac{\partial\rho}{\partial t}. \quad (80)$$

The focus of this formulation is the first term on the right-hand side, which involves the rate of change of the volumetric water content. This term accounts for both the change in saturation and the change in porosity, with the latter resulting from the compressibility of the porous matrix. The second term on the right-hand side accounts for the compressibility of water.¹¹

The first term on the right-hand side of equation 80 is the only one considered in Celia’s work. Thus, Celia’s approach is based on the assumption that the water is incompressible ($\frac{\partial\rho}{\partial t} = 0$), which makes the second term on the right-hand side vanish. However, note that if we further simplify the expression by also assuming confined conditions ($\frac{\partial s}{\partial t} = 0$), then equation 80 reduces to

$$\frac{\partial(\rho s\phi)}{\partial t} = \rho s \frac{\partial\phi}{\partial t}. \quad (81)$$

This expression accounts only for the compressibility of the porous matrix. While this might be appropriate in some use cases, it is not generally what we expect in terms of accounting for compressibility effects (which we typically want to include water), i.e., specific storage. And yet, from this analysis we see that it is inherent in the Celia-type approach based on volumetric water content.

According to Farthing and Ogden (2017) [need a proper reference here], it is common to account for specific storage by simply adding it to the Celia-type storage term. However, because the Celia term already accounts for the compressibility of the porous matrix, adding the full specific storage term results in “double counting” the effect of porous matrix compressibility. A proper accounting of specific storage could presumably be achieved by a “modified Celia-type” approach that includes the density-derivative term (the second term on the right-hand side of equation 80, instead of the full specific storage term) and tracks the evolution of density with changing pressure via the water compressibility. That said, it is worth noting that adding the full specific storage term might not be so bad because in practice the Celia-type term, when related to a water retention curve measured in the laboratory, might not actually account for matrix compressibility effects found in the field. In that case, one can pull ϕ out of the derivative in the Celia-type term, resulting in the saturation-based formulation we discuss next.

¹¹It might also be considered to include the effects of solute concentration and temperature on density, but that’s another topic altogether, and a rather deep one.

B.2 Formulation in terms of saturation

Another alternative is to separate out only the saturation-derivative term, leaving the product $\rho\phi$, the mass of water per volume of aquifer (“bulk density of water”) at full saturation, intact as a dependent variable:

$$\frac{\partial(\rho s \phi)}{\partial t} = \rho\phi \frac{\partial s}{\partial t} + s \frac{\partial(\rho\phi)}{\partial t} \quad (82)$$

The focus of this formulation is the first term on the right-hand side, which involves the rate of change in saturation. The second term on the right-hand side accounts for the compressibility of the water and the porous matrix, i.e., specific storage.

This is the formulation we’ve adopted in this writeup, and it is consistent with the saturation-based formulations used in VS2D [5] and SUTRA [11]. To conserve mass, it appears we need to either track the evolution of the specific storage term with time (see equation 58) or neglect specific storage altogether when the medium is partially saturated. Tracking specific storage should provide a “smooth” transition between partially and fully saturated conditions, as the full specific storage term would always be accounted for.¹² Neglecting specific storage under partially saturated conditions would obviously cause a “discontinuity” in that that term would suddenly come into play under under fully saturated conditions. Not sure it makes a whole lot of difference from a model accuracy standpoint. Not tracking specific storage would certainly be simpler.

B.3 Formulation in terms of “mass saturation”

The final alternative is to separate out only the porosity-derivative term, leaving the product ρs , the mass of water per volume of pores (“mass saturation” or “pore density of water”¹³), intact as a dependent variable:

$$\frac{\partial(\rho s \phi)}{\partial t} = \phi \frac{\partial(\rho s)}{\partial t} + \rho s \frac{\partial\phi}{\partial t} \quad (83)$$

The focus of this formulation is the first term on the right-hand side, which involves the rate of change of the “mass saturation”. This term accounts for both the change in saturation and the change in density, with the latter resulting from the compressibility of the water. The second term on the right-hand side accounts for the compressibility of the porous matrix.

Here, one is apparently faced with a choice between either tracking the evolution of porosity or assuming it is constant, i.e., that the porous matrix is incompressible. If porosity is constant, this formulation suffers from a limitation similar to that of the Celia-like formulation. Under confined conditions ($\frac{\partial s}{\partial t} = 0$),

¹²Would we need to continue tracking specific storage explicitly even under fully saturated conditions? I think not, but I’m not entirely sure.

¹³Terms I made up; not used by anyone else as far as I know.

the expression reduces to

$$\frac{\partial(\rho s \phi)}{\partial t} = s \phi \frac{\partial \rho}{\partial t}. \quad (84)$$

which considers only the compressibility of the water and thus does not allow us to account fully for specific storage. Another issue with this formulation is that there probably are no parameterizations available for calculating ρs as a function of pressure or pressure head.

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